Stale Rollouts Can Pick the Wrong Prompts

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Abstract

Reinforcement learning with verifiable rewards (RLVR) increasingly ranks training prompts with *stale* rollouts—responses generated by an earlier policy—after a one-sided importance correction: prefix-ratio methods restore where the current policy goes, future-trace methods restore how it finishes, and token-ratio influence scores restore neither. We ask when such stale selection can be trusted, and answer in three tiers. **(i) Possibility (proved).** We organize existing estimators into a 2×2 decomposition over prefix occupancy and continuation outcome and prove that any one-sided correction can reverse the sign of a prompt's policy gradient at arbitrarily small policy divergence; the failure survives group normalization, and KL, ESS, and pointwise-cosine dashboards provably cannot exclude it. Cell *disagreement* is the one stale-computable warning sign: in a seeded random search over $5{\times}10^4$ two-token environments, every double reversal was accompanied by disagreement between the two one-sided estimators. **(ii) Realization map (measured).** Across Qwen2.5 7B/14B and three math tasks under controlled LoRA drift, whether reversal materializes depends on the signal regime, which we measure by a corrected split-half oracle floor. The only regime with measurable ranking signal (floor 0.32 vs. chance 0.10) shows one-sided precision loss, while in weak-signal regimes stale scores track the oracle (Fisher-combined exact $p = 1.0{\times}10^{-6}$), and on a saturated task the selection problem itself degenerates (floor 0.080, below chance). **(iii) Impossibility (proved + measured).** Two impossibility results close the loop. First, certifying the top-k decision is out of reach: alignment margins at the ranking boundary (0.00–0.24°) sit *5–8 orders of magnitude in rollout budget* below achievable confidence radii, across an 8× drift range and all selection fractions 5–25%. Second, we prove a *measurement ceiling*: the precision of *any* estimator measured against a K-rollout oracle is bounded by a function of the split-half floor, so when the floor is near chance no method can measurably win—our own stale baseline already sits at this ceiling (0.294 observed vs. ≈0.30 ceiling). We therefore contribute a GPU-free diagnosis procedure that decides, from existing artifacts alone, whether a prompt pool supports selection at all (conditional-subset precheck plus a K-scaling floor curve with Spearman–Brown extrapolation), and report its pre-registered verdict on our pools: structural absence. Along the way we document four evaluation pitfalls that our own pipeline caught—an oracle-sample reuse leak, shared-jitter tie inflation, cell-budget imbalance, and saturation-shared overlap inflation—each with a guard. The practical prescription is *measure the floor first*: selection, and even the evaluation of selection, is only meaningful where recoverable ranking signal exists, and today's dashboards do not tell you.

1 Introduction

RLVR pipelines score each candidate prompt by "how much would training on it help the current policy π" and keep the top k. Exact scores require fresh rollouts from π for every candidate at every selection round, which is expensive; practice therefore reuses rollouts and rewards produced by an

*Draft v0.1 (2026-08-13). Reframed structure following the pre-registered "structural absence" branch; numbers are from completed runs (v1 gate + v2 corrected, 2 seeds); pending seed harvests will update Sections 5 and 8 only.

earlier behavior policy β, together with an importance correction. The correction has two axes: (a) how the current policy *reaches* a token (prefix occupancy) and (b) how the current policy *finishes* after it (continuation outcome). Existing estimators correct exactly one axis—not by oversight, but because correcting both multiplies hundreds of token ratios and explodes the variance. The implicit bet behind this truncation is that the remaining half induces a small bias that roughly preserves *ranking*.

This paper audits that bet end to end and reports that the situation is worse, and stranger, than the bias–variance framing suggests. Our claims come in three tiers, in decreasing order of generality and increasing order of empirical content:

1. **Possibility (proved).** One-sided corrections can reverse the sign of a prompt's gradient at arbitrarily small policy divergence (Section 3). Group normalization does not rescue the sign, and the dashboards practitioners watch—KL bounds, ESS, pointwise gradient cosine—provably cannot exclude the failure. The one stale-computable warning sign is *disagreement between the two one-sided cells.*

2. **Realization map (measured).** Whether reversal materializes depends on the task–capability regime (Section 5). We chart both directions with a pre-registered protocol: a regime where one-sided selection measurably loses precision, a regime where stale selection provably tracks the oracle, and a regime where the selection problem degenerates outright.

3. **Impossibility (proved + measured).** Where selection matters most, neither the decision nor its evaluation can be secured at practical budgets: boundary certification requires 5–8 orders of magnitude more rollouts than practical (Section 6), and a measurement ceiling caps what any estimator can even *demonstrate* against a noisy oracle (Section 4).

The empirical protagonist of the paper is not an estimator but a measurement: the *split-half oracle floor*, the top-k agreement between two independent halves of the oracle's own rollouts. The floor measures how much prompt-ranking signal is recoverable at a given rollout budget; our measurement ceiling result (Proposition 2) makes it the quantity that decides whether comparative claims about selection methods are meaningful at all. Applying our own medicine, we ran a pre-registered, GPU-free diagnosis on our prompt pools (Section 8) and obtained the verdict *structural absence*: on saturated or tie-degenerate pools, no increase of the oracle budget up to $K{=}256$ rollouts per prompt is predicted to create a signal regime. That verdict, rather than a victory lap for a new estimator, is the paper's empirical headline—and, we argue, the more useful contribution for practitioners deciding where their fresh-rollout budget should go.

A final theme runs through the experimental sections. Selection evaluation is easy to get silently wrong in the optimistic direction: over the course of this project our own pipeline produced—and then caught—four distinct evaluation artifacts, each of which had initially manufactured a publishable-looking positive result (Section 7). We document each mechanism and its guard, because we believe they are endemic to the growing literature on rollout reuse and data selection for RLVR.

2 Related work

Off-policy influence and prompt selection. Token-ratio influence scores rank prompts from offline trajectories with a single-token correction [CROPI]; in our decomposition this is g_{00} (neither axis restored). Its own appendix already shows the symptom that motivates our estimand: high pointwise gradient cosine (40/50 prompts above 0.6) coexisting with only 28.8% top-10% neighborhood preservation. Prefix-ratio corrections [A Step Back, Cumulative] restore occupancy (g_{10});

their guarantees assume a target-policy advantage that practical critic-free objectives replace by a behavior-generated outcome reward—precisely the gap our counterexamples exploit. Future-trace corrections [Multi-Step] restore the continuation (g_{01}) but not occupancy when used for per-prompt gradient ranking. Full-trajectory corrections [TIC-GRPO] are exact but variance-explosive on long responses; they serve as the cost anchor, not a practical selector.

Rollout allocation and learning stability. Allocation methods [GradAlign, LearnAlign] decide where to spend fresh rollouts to maximize training yield, and stale-data stabilizers keep *learning* well-behaved under reuse; neither certifies, or even measures, the correctness of the top-k *selection decision*, which is the object practitioners actually consume. Classical off-policy correction and subset selection give the statistical backdrop: conditional importance sampling [Rowland et al.], stationary-distribution correction [Liu et al.], doubly-robust policy gradients [Huang & Jiang], and information-complexity lower bounds for top-k identification [Kaufmann et al.]. Our certification impossibility (Section 6) is an empirical instance of the latter's warning: when value gaps at the boundary vanish, identification cost diverges.

Reliability of noisy rankings. Our floor and ceiling analysis imports split-half reliability and the Spearman–Brown prophecy [Spearman, Brown] into RLVR selection evaluation. To our knowledge, no prior work in this literature reports the reliability of its own oracle, and Section 4 shows several of our initially-promising comparisons dissolve once it is reported.

3 Theory: one-sided corrections and what dashboards cannot see

3.1 A 2×2 decomposition of stale-gradient estimators

For prompt x, response $Y = (A_1, \ldots, A_T)$ with terminal verifiable reward $R(Y)$, the current-policy gradient is $g_\pi(x) = \sum_t \mathbb{E}_{H_t, A_t \sim \pi}[Q_t^\pi(H_t, A_t)\, z_t]$ with $z_t = \nabla_\theta \log \pi(A_t \mid H_t)$. With behavior trajectories, token ratios $r_j = \pi(A_j|H_j)/\beta(A_j|H_j)$, prefix products $P_t = \prod_{j<t} r_j$ and suffix products $S_t = \prod_{j>t} r_j$, four population quantities arise from the same stale data:

$$
\begin{aligned}
g_{00} &= \textstyle\sum_t \mathbb{E}_\beta[r_t R z_t] && \text{(occupancy } \beta\text{, continuation } \beta\text{)}\\
g_{10} &= \textstyle\sum_t \mathbb{E}_\beta[P_t r_t R z_t] && \text{(occupancy } \pi\text{, continuation } \beta\text{)}\\
g_{01} &= \textstyle\sum_t \mathbb{E}_\beta[r_t S_t R z_t] && \text{(occupancy } \beta\text{, continuation } \pi\text{)}\\
g_{11} &= \textstyle\sum_t \mathbb{E}_\beta[P_t r_t S_t R z_t] = g_\pi.
\end{aligned}
$$

The subscripts state which axis is restored. The exact error decompositions

$$
g_{10} - g_\pi = \sum_t \mathbb{E}_{d_\pi,\pi}\big[(Q_t^\beta - Q_t^\pi) z_t\big] \quad \text{(continuation bias)}, \qquad g_{01} - g_\pi = \sum_t \big(\mathbb{E}_{d_\beta,\pi} - \mathbb{E}_{d_\pi,\pi}\big)\big[Q_t^\pi z_t\big] \quad \text{(occupancy bias)}
$$

show that the two one-sided estimators remove *different* biases; they are not substitutes.

3.2 Sign reversal at arbitrarily small divergence

Theorem 1 (One-sided sign reversal)**.** *For every $\epsilon > 0$ there exist two-token environments and policy pairs (π, β) with trajectory KL $O(\epsilon^2)$ such that $g_\pi = -\epsilon/2$ while $g_{00} = g_{10} = +\epsilon/2$ (continuation counterexample), and likewise $g_{00} = g_{01} = +\epsilon/2$ (occupancy counterexample). Group normalization with any group size $K \geq 2$ preserves the reversal.*

The two constructions are given in Appendix A; both flip a Bernoulli success profile so that the policies agree wherever the retained correction looks and disagree exactly where the discarded half lives. Both are machine-checked (raw gradients, exact enumeration of group normalization for $K \in \{2, 4, 8\}$ and $\epsilon \in \{0.4, 0.1, 0.01\}$) by a self-contained verification script shipped with the code.

Corollary 1 (Dashboard blindness)**.** *No bound on trajectory KL, no lower bound on ESS, and no norm-blind, scale-invariant single-estimator diagnostic (e.g. pointwise cosine of the estimator against itself across checkpoints) can exclude sign reversal of a prompt's selection score: in the constructions above all such quantities are driven to their benign limits while the ranking direction is reversed. A gradient-norm floor or a selection-margin bound is necessary; and since $\epsilon \to 0$ also drives the absolute gradient to zero, normalized cosines conceal exactly the missing signal floor.*

Proposition 1 (Disagreement accompanies double failure)**.** *In the continuation counterexample, $\cos(g_{10}, g_{01}) = -1$: the two one-sided cells disagree maximally. In a seeded random search over 5×10^4 two-token, two-parameter environments we found* zero *instances of a double reversal (both one-sided estimators reversed against g_π) accompanied by cell agreement ($\cos(g_{10}, g_{01}) > 0.9$); the identity $g_{11} \equiv g_\pi$ was verified on every sample.*

Proposition 1 is deliberately modest: absence of disagreement is *not* proven to imply safety in general architectures with parameter sharing, disagreement cannot certify *which* cell is right, and its practical resolution is limited by the same margin collapse quantified in Section 6. We therefore use cell disagreement only as an *unreliability signal*—a cheap, stale-computable reason to distrust a ranking—never as a selector. The mechanism is transparent from the error decompositions: g_{10} and g_{01} subtract different bias terms from the same truth, so a large simultaneous failure of both requires the two bias terms to agree, which the decomposition makes non-generic.

4 Measuring selection: floor, ceiling, and when comparison is meaningful

Estimand. All experiments target the decision practitioners consume: the top-k set by $s_i = \cos(\mu_i, v)$, where μ_i is the prompt's mean current-policy gradient and v a validation mean gradient. We report top-k precision against an on-policy oracle, boundary margins, and rollout budgets. Pointwise cosine averages are diagnostics, never headlines.

The split-half floor. The oracle itself is a K-rollout Monte Carlo estimate. We therefore report, for every run, the *corrected split-half floor*: rollouts are split into two disjoint halves (odd/even micro-groups), each half induces a top-k set (ties broken by *independent* per-half jitter; Section 7 explains why this correction matters), and the floor is the mean overlap fraction over 20 jitter pairs. Chance is k/n. The floor measures recoverable ranking signal at budget K; it is conservative (an estimator can measurably exceed it), and Section 7 shows it is itself vulnerable to tie artifacts unless computed exactly this way.

A measurement ceiling. The floor does more than describe the oracle: it bounds what any comparison against that oracle can show.

Proposition 2 (Measurement ceiling)**.** *Model prompt scores as exchangeable Gaussians: true value t_i, half-oracle scores a_i, b_i with $\mathrm{corr}(a, b) = \rho_h$, full-$K$ oracle score o_i with reliability $\rho_f = 2\rho_h/(1+\rho_h)$ (Spearman–Brown). Then for* any *estimator s computed independently of the oracle's sampling noise,*

Table 1: Floor → ceiling mapping under Proposition 2 (n=512, k=51, chance 0.100; simulation, 600–800 replicates/cell). The v2 weak-signal regime (floors 0.137–0.176) caps *any* method's measurable precision at ≈0.30–0.37; the observed stale baseline already sits there.

observed floor	0.104	0.113	0.130	0.139	0.170	0.262	0.383
half-reliability ρ_h	0.02	0.05	0.10	0.12	0.20	0.40	0.60
ceiling on measurable precision	0.170	0.219	0.278	0.300	0.372	0.516	0.640

the expected top-k overlap between s and the K-rollout oracle is at most that of the best possible estimator $s = t$*, namely* $\mathrm{Ovl}(\sqrt{\rho_f})$*, where* $\mathrm{Ovl}(c)$ *is the expected top-k overlap of two Gaussian scores with correlation c. The observable floor equals* $\mathrm{Ovl}(\rho_h)$*, so the ceiling is a computable, monotone function of the observed floor.*

Table 1 gives the mapping at our experimental geometry (n=512, k=51), validated by simulation (Appendix B). Two consequences organize everything that follows. First, when the floor is near chance the ceiling collapses toward chance: *no estimator can measurably win, and comparative evaluation of selection methods is void.* Second, an estimator observed *at* the ceiling has plausibly exhausted the measurable signal—further method development cannot show gains without first raising the floor (more oracle rollouts, or a pool with more signal).

Retention and its domain. When comparing regimes we report retention $(\mathrm{prec} - c)/(\mathrm{floor} - c)$ with c the chance level, declared *undefined* whenever $\mathrm{floor} - c \leq 0.02$; saturated cells otherwise produce sign-flipping or exploding ratios (the 14B GSM8K cell below is the cautionary case).

5 The realization map

Setup. Qwen2.5-7B/14B-Instruct; GSM8K, MATH-500, and DAPO-Math-17k; drift between β and π is controlled by LoRA RFT for 50–400 steps; gradients are JL-projected; oracle uses K=32 fresh rollouts per prompt (8 micro-groups × 4); pools are n=256 (v1) or n=512 (v2) with k=10%. The v2 protocol was pre-registered after a full audit (equal per-cell budgets, seeded tie-breaks, independent jitter, manifests); judgment criteria were fixed before results were read. Two v2 seeds per task have completed at the time of writing; remaining seeds update numbers, not verdicts, and the pre-registered analysis will be re-run on arrival.

5.1 Where signal exists, one-sided selection can lose it

The v1 gate (7B GSM8K, floor 0.32—the strongest signal regime we ever measured) passed its pre-registered criteria: one-sided estimators lose top-10% precision against the uncorrected baseline, and at the largest drift (400 LoRA steps) the prefix-corrected estimator collapses to precision 0.040 against a floor of 0.320 (overlap 1/25; exact hypergeometric $P(X \leq 1) = 0.27$, so we flag this as a directional observation, not an individually significant below-chance claim). The misranking propagates: selecting by one-sided scores changes downstream fine-tuned accuracy (base 0.740: oracle-selected 0.840 vs. one-sided 0.840/0.880 vs. random 0.820)—though in this saturation-adjacent regime the spread is small, foreshadowing what follows. On 7B MATH-500 (floor 0.320) the *ordering flips*: the uncorrected g_{00} is worst (0.160) and every stale estimator degrades broadly (0.28–0.64 of available signal); at 14B MATH-500 (floor 0.240) the loss concentrates on the suffix axis (signal

Table 2: v2 corrected results (n=512, k=51, chance 0.100; 2 seeds). Floors are corrected split-half (independent jitter). DAPO precisions are jitter-sensitive (±0.08) due to margin-0 tie degeneracy and shown as bands.

run	floor	g_{00}	g_{10}	g_{01}	g_{11}	regime
GSM8K s1	0.137	0.294	0.275	0.196	0.196	weak signal; stale at ceiling
GSM8K s2	0.176	0.235	0.196	0.216	0.216	weak signal
DAPO s1	0.176	0.04–0.16 (tie-degenerate)				margin 0.0; chance
DAPO s2	0.176	0.04–0.16 (tie-degenerate)				margin 0.0; chance

retention g_{01}=0.44 vs. g_{10}=1.00). Which cell fails is regime-dependent; *that some cell fails while dashboards stay green* is the invariant, exactly as Theorem 1 licenses and no more.

5.2 Where signal is weak, stale selection tracks the oracle

The corrected v2 runs put every regime we have at floor 0.137–0.176 against chance 0.100 (Table 2). Here stale scores are *not* distinguishable from the oracle—in fact they track it: on GSM8K the uncorrected stale baseline achieves precision 0.294/0.235 (seeds 1/2), upper-tail exact $p = 2.8 \times 10^{-5}$ and 2.1×10^{-3}, Fisher-combined $p = 1.0 \times 10^{-6}$. Crucially, 0.294 is *at the measurement ceiling* (≈0.30 at floor 0.137; Table 1): the stale estimator has exhausted what the oracle can resolve, so neither its one-sided competitors' losses nor any method's gains are measurable in this regime. The DAPO pool is tie-degenerate (boundary margins exactly 0.0; precision itself moves by ±0.08 under tie-break jitter, so we report bands): all stale estimators sit at chance, and so does everything else.

5.3 Where capability saturates, selection degenerates

At 14B on GSM8K (∼95% accuracy) the pool contains 132/256 zero-reward-variance prompts; every estimator's precision equals the chance level (0.087–0.111 vs. 0.098), and the oracle disagrees *with itself* below chance (floor $0.080 < 0.098$). Selection, ranking, and certification all degenerate together: no method, and no certificate, can matter here. The practical corollary is that selection pipelines must scale candidate difficulty with model capability, or silently pay for machinery that cannot change the outcome.

6 Certification is out of reach where it matters

Even where a point estimate selects well, one may ask for the decision to be *certified*: spend fresh rollouts near the ranking boundary until confidence intervals separate the top-k set. We implemented this (sequential boundary refinement with shared-validation error propagation; pseudocode in Appendix C) not as a method contribution but as a measuring instrument, and it returns a structural negative: across every completed condition (both scales, three tasks, drift 50–400, selection fractions 5–25%) certification fails at budgets 1.02–1.04× the uncorrected cost, because top-k boundary margins are 0.00–0.24° while achievable confidence radii are 51–180°. The gap is not a constant factor: with radius $\alpha_v(m) = \arcsin(c/\sqrt{m})$, reaching the observed margins requires $\times 4.1{\times}10^5$ to $\times 1.0{\times}10^8$ more observations—5–8 orders of magnitude in budget, robust to (ϵ, δ)-PAC slack and deepened validation. Prompt value is intrinsically dense at the selection boundary: *selection decisions cannot be certified precisely where selection matters.* Fresh rollouts spent on certification are wasted; Section 9 argues they belong in floor measurement instead.

7 Four evaluation pitfalls our own pipeline caught

Every mechanism below manufactured, at some point in this project, a clean positive result that survived a first review and died under a targeted audit. We report them as first-class findings; each comes with a guard now built into our protocol, and we suspect instances of each exist in the current rollout-reuse literature.

P1: Oracle-sample reuse (leakage). Our v1 hybrid intervention (completing the "missing half" with fresh continuations) showed a perfect 21/21-condition recovery, mean +0.46/+0.49 precision—the would-be causal centerpiece. A code audit found the treated cell reused rollouts that also defined the oracle; the shared noise, not causation, produced the recovery. A pre-registered equal-budget decisive experiment with disjoint samples showed *no consistent recovery* (e.g. GSM8K s1: fully-fresh cell 0.38 *below* mixed cell 0.46). *Guard:* strict sample disjointness between treatment and oracle (odd/even micro-group separation), enforced structurally.

P2: Shared-jitter tie inflation. Our first floor implementation broke score ties with a jitter shared across the two halves. On a tie-degenerate pool (DAPO, margins 0.0) this inflated the floor from 0.176 to 0.804—which for a week supported a dramatic (and wrong) "inverse correlation" headline. *Guard:* independent per-half jitter streams, 20 seeded pairs, plus reporting precision *bands* under jitter on tie-heavy pools.

P3: Cell-budget imbalance. Comparing estimator cells with unequal effective rollout counts silently favors the better-funded cell; our v1 hybrid grid had this flaw on top of P1. *Guard:* equal-K redesign of all cell comparisons, verified from manifests.

P4: Saturation-shared overlap inflation. An above-chance floor can be an artifact of *shared liveness* rather than shared ranking: prompts that saturate in both halves (all-correct or all-wrong) drop to an exactly-tied score, and the surviving "live" sets overlap between halves, producing top-k agreement with zero true ranking signal. In a synthetic pool with *no* ranking signal, saturation sharing alone yields floor 0.126 at K'=8, decaying toward chance (0.110) at K'=32—the signature being a floor that *decreases* as oracle budget grows. Our measured K-scaling curves (Section 8) show exactly this decreasing shape, which means even our reported weak-signal floors are best read as upper bounds. *Guard:* always report the floor as a function of K'; treat a non-increasing floor curve as artifact evidence, and report tie mass alongside.

8 Diagnosing signal absence without GPUs

The pitfalls motivate the final contribution: a pre-registered, GPU-free procedure that decides whether a pool supports selection *before* any new experiment is funded. It has two stages, both computed from artifacts every RLVR selection run already produces.

Stage 1: conditional-subset precheck. Any proposed pool intervention definable as a filter on existing prompts (e.g. "keep only live prompts, $0 <$ pass-rate < 1") already exists as a subset of completed runs. Computing the corrected conditional floor on that subset (with bootstrap CIs) prechecks the intervention at zero GPU cost: if the conditional floor does not clear $2\times$ chance in a majority of eligible runs, building the new pool has no evidential basis. On our pools this

Table 3: K-scaling floor curves (observed exact recomputation; prediction by Spearman–Brown + Gaussian overlap). Chance 0.100. Note the *decreasing* observed curves—the P4 artifact signature—and the near-zero half-reliabilities. Verdict (pre-registered): structural absence.

run	r_1	K'=8	16	24	32	pred. 128	pred. 256
GSM8K s1	0.045	0.263	0.208	0.201	0.182	0.288	0.391
GSM8K s2	−0.005	0.220	0.182	0.170	0.156	0.094	0.094
DAPO s1	0.009	0.112	0.123	0.139	0.146	0.147	0.182
DAPO s2	−0.038	0.119	0.148	0.158	0.158	0.094	0.094

returned NO-GO, and the intervention (a live-only GSM8K pool) was cancelled without spending a GPU-hour.

Stage 2: K-scaling floor curve. From stored micro-group gradients the floor is *exactly* recomputed at $K'=8, 16, 24, 32$, and extrapolated to K'=64–256 by Spearman–Brown on the half-reliability r_1 mapped through the Gaussian overlap function, with a calibration check at the largest observed K'. The pre-registered verdict rule: if a majority of eligible runs are predicted to reach 2× chance by $K' \leq 256$, recommend oracle expansion; otherwise declare structural absence.

Table 3 shows the result: one GSM8K seed extrapolates to 2× chance at K'=128 from r_1=0.045; the other is flat at chance; both DAPO runs never reach it. Majority not met: *structural absence.* Two honest caveats are pre-registered with the method. The calibration check shows the Gaussian model under-predicts observed floors at K'=32 (e.g. 0.094 predicted vs. 0.156 observed)—consistent with the P4 artifact inflating the *observed* values, which makes the verdict conservative in the safe direction. And the single positive extrapolation rests on an r_1 small enough to be within the P4 artifact's reach, so we treat it as motivation for a cheap re-check when further seeds land, not as grounds for a GPU expansion.

9 Practical guidance, limitations, and outlook

For practitioners. (1) *Measure the floor first.* A 48-prompt fresh probe suffices to estimate the split-half floor; below 2× chance (a threshold to fix *before* looking), retire selection and spend nothing—random selection is free and provably indistinguishable. (2) *Never budget for boundary certification*; the 5–8 order gap is structural. (3) *Scale candidate difficulty with capability*, or selection degenerates regardless of estimator. (4) *Report floor and ceiling with any selection comparison*; a claimed improvement above the measurement ceiling of its own oracle is an artifact by construction. (5) *Distrust one-sided rankings when the two one-sided cells disagree*; agreement guarantees nothing, but disagreement is a free warning.

Limitations. Empirics are Qwen-2.5 (7B/14B), math tasks, and LoRA drift; two completed v2 seeds (remaining seeds and a small-model capability–difficulty sweep are running and will be folded in; the verdict logic is pre-registered). The ceiling proposition assumes an exchangeable Gaussian score model (simulation-validated at our geometry; the analytic program of proving it distribution-free is open). The disagreement result is an existence-plus-search finding, not a general sufficiency theorem. Theory counterexamples are two-token constructions—deliberately: they show what *cannot* be excluded by dashboards, not what typically happens; the realization map is the empirical complement.

Outlook. We view the negative results as load-bearing: they delimit where the field's growing investment in stale-rollout selection can possibly pay off (unsaturated, tie-free, signal-bearing pools—which our diagnosis identifies for free) and where it cannot. The measurement-first protocol—floor, ceiling, preregistration, and the four guards—is applicable to any selection-method paper, including ours.

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A Counterexample constructions and machine verification

Continuation counterexample (reverses g_{00}, g_{10}). Two-token MDP; first action $A_1 \sim \text{Bernoulli}(q)$, q=1/2, identical under both policies; reward $R = A_2$. Behavior success profile $Q^\beta(A_1{=}0) = \frac{1}{2} - \epsilon$, $Q^\beta(A_1{=}1) = \frac{1}{2} + \epsilon$; current policy swaps the two values. The gradient w.r.t. the logit of q is $g_\pi = -\epsilon/2$ while $g_{00} = g_{10} = +\epsilon/2$: the prefix already matches perfectly, so a perfect prefix ratio changes nothing, and the behavior-policy *ending* drives the sign. Mean suffix KL is $O(\epsilon^2)$; the 1-D cosine is -1 throughout.

Occupancy counterexample (reverses g_{00}, g_{01}). Behavior visits $P(A_1{=}0) = \frac{1}{2} + \epsilon$, current $\frac{1}{2} - \epsilon$; the second action is Bernoulli(1/2) under both, and $R = A_2$ if $A_1{=}0$, else $1 - A_2$. For the second-action logit, $g_\pi = -\epsilon/2$ while $g_{00} = g_{01} = +\epsilon/2$: no tokens remain after A_2, so the future ratio is exactly 1, and the stale *visit* distribution drives the sign. Trajectory KL is again $O(\epsilon^2)$.

Group normalization. Both constructions have overall success rate 1/2 under both policies; binary group standardization multiplies each direction by the same nonnegative scalar, hence preserves the reversal (exact enumeration for $K \in \{2, 4, 8\}$, $\epsilon \in \{0.4, 0.1, 0.01\}$ in the verification script, alongside leave-one-out unbiasedness and the confidence radius bound used in Section 6).

B Measurement ceiling: derivation sketch and validation

Under the model of Proposition 2, write the full-K oracle score as $o = \sqrt{\rho_f}\, t + \sqrt{1 - \rho_f}\, \eta$ with η independent noise. For any estimator s independent of η, $\mathrm{corr}(s, o) = \mathrm{corr}(s, t)\sqrt{\rho_f} \leq \sqrt{\rho_f}$, with equality iff s is (a monotone transform of) t. Expected top-k overlap between two jointly Gaussian scores is increasing in their correlation, which yields the ceiling $\mathrm{Ovl}(\sqrt{\rho_f})$; the observable floor is $\mathrm{Ovl}(\rho_h)$ by the same function. Simulation at $n{=}512$, $k{=}51$ (600–800 replicates per cell; independent-jitter tie handling) produces Table 1; the mapping is monotone and smooth, so inverting the observed floor through it is well-posed. The Gaussian assumption matters: heavy-tailed score distributions with a stable head can exceed the Gaussian ceiling estimate, which is precisely the P4 caveat and why we report the ceiling alongside, not instead of, the K-scaling curve.

C Sequential boundary certification (measuring instrument)

Off-policy scores serve as a prior; every candidate receives a small fresh micro-group; scores carry confidence balls (shared validation error propagated); fresh rollouts are then spent only on candidates whose balls straddle the top-k boundary, until $\min L_{\text{selected}} > \max U_{\text{rest}}$ or the budget is exhausted, in which case *certification failure is reported, never hidden.* Variants (scalar CI, (ϵ, δ)-PAC with $\epsilon \leq 0.05$) behave identically in all our conditions because the binding constraint is the boundary margin itself (Section 6).

D Preregistration and audit trail

All v2 judgment criteria, the precheck GO/NO-GO rule, the K-scaling verdict rule, and the reframing decision tree ("structural absence $\Rightarrow$ this paper's structure") were registered in the project log before the corresponding results were computed; the four pitfalls each carry a before/after commit pair. The verification script, manifests, and all readouts ship with the code release.